Transport in mammals – AS 9700 November 2023

1. Nov/2023/Paper_ 9700/11/No.30

Which factors affect blood pressure?

1 the diameter of the blood vessels

2 the systolic pressure of the heart ventricles

3 the volume of blood returning to the heart in each cardiac cycle

A 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 2 and 3 only

2. Nov/2023/Paper_ 9700/11/No.32

The partial pressure of oxygen in blood vessels in different parts of the body affects the percentage saturation of each haemoglobin molecule with oxygen.

blood vessel	partial pressure of oxygen / kPa	saturation of each haemoglobin molecule with oxygen / %
vein in active muscle	4	25
vein in resting muscle	7	75
arteriole in lungs	12	100

Which statement explains the difference in saturation between a resting and an active muscle?

A A small decrease in partial pressure causes two atoms of oxygen to leave each haemoglobin molecule because the oxygen dissociation curve is sigmoid shaped.

B A small decrease in partial pressure between 4 kPa and 7 kPa causes two molecules of oxygen to leave each haemoglobin molecule because the oxygen dissociation curve is steep.

C Loading of oxygen is slower between 4 kPa and 7 kPa due to binding of one oxygen molecule to each haemoglobin molecule making the next one harder to bind.

D Loading of oxygen is faster between 4 kPa and 7 kPa due to binding of one oxygen molecule to each haemoglobin molecule making the next one easier to bind.

3. Nov/2023/Paper_ 9700/11/No.34

Some of the events during a cardiac cycle are listed.

1 Left and right atria contract.

2 Left and right ventricles contract.

3 A wave of electrical activity is conducted by the Purkyne tissue.

4 A wave of electrical activity passes through the atrioventricular node.

What is the correct sequence of events after the sinoatrial node sends out a wave of electrical activity?

A $1 \rightarrow 3 \rightarrow 4 \rightarrow 2$

B $1 \rightarrow 4 \rightarrow 3 \rightarrow 2$

C $2 \rightarrow 4 \rightarrow 3 \rightarrow 1$

D $4 \rightarrow 1 \rightarrow 3 \rightarrow 2$

4. Nov/2023/Paper_ 9700/12/No.26

Which processes are required for the transfer of assimilates from a source cell into a phloem sieve tube?

A active transport, facilitated diffusion and simple diffusion

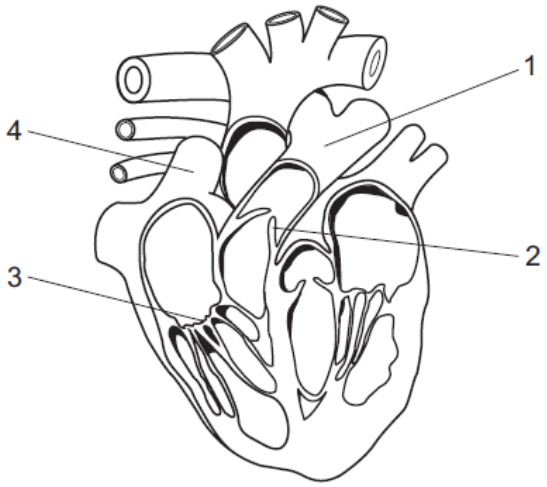
B active transport and exocytosis

C active transport and simple diffusion only

D active transport only

5. Nov/2023/Paper_ 9700/12/No.29

The diagram shows an anterior view of a vertical section through the human heart.



Which row shows the correct labels for the diagram?

	1	2	3	4
A	aorta	atrioventricular valve	semilunar valve	pulmonary artery
B	pulmonary vein	semilunar valve	atrioventricular valve	pulmonary vein
C	pulmonary artery	semilunar valve	atrioventricular valve	vena cava
D	pulmonary vein	atrioventricular valve	semilunar valve	vena cava

6. Nov/2023/Paper_ 9700/12/No.30

What does the inner layer of veins **and** capillaries contain?

A collagen fibres

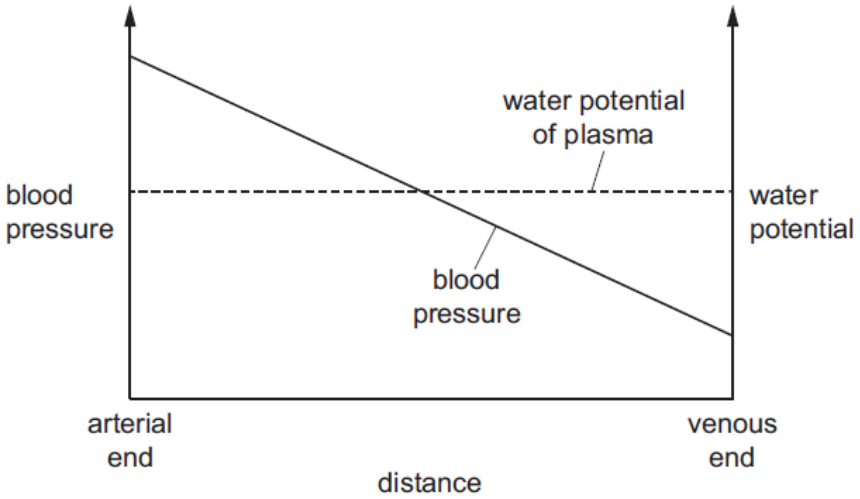
B elastic fibres

C endothelium

D smooth muscle

7. Nov/2023/Paper_ 9700/12/No.31

The graph shows the changes in blood pressure and water potential across a blood capillary from the arterial end of the capillary to the venous end of the capillary.



What could result in an accumulation of excess tissue fluid?

A an increase in concentration of large proteins in the blood plasma

B an increase in blood pressure

C a decrease in the concentration of small proteins in tissue fluid

D a decrease in blood pressure

8. Nov/2023/Paper_ 9700/12/No.32

Which statement is correct about a red blood cell that has just entered a capillary in a respiring muscle?

A There is a net decrease in chloride ions.

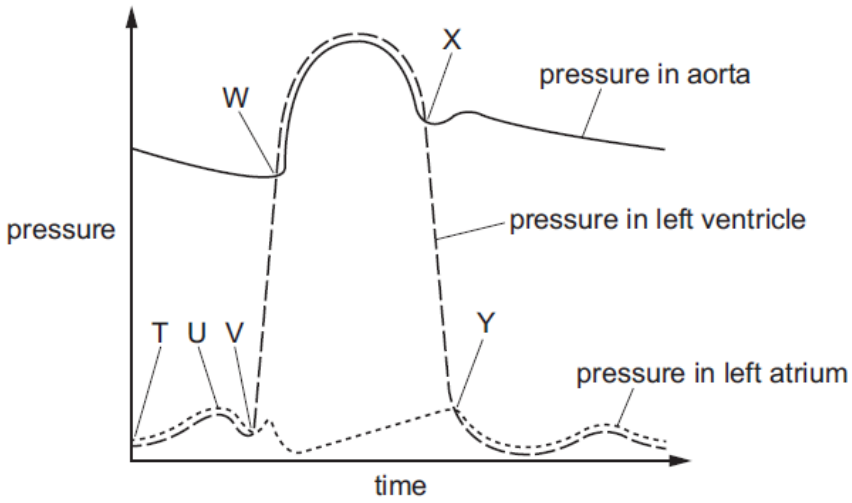
B There is a net decrease in haemoglobinic acid.

C There is a net increase in carbonic anhydrase.

D There is a net increase in carbaminohaemoglobin.

9. Nov/2023/Paper_ 9700/12/No.33

The graph shows some blood pressure changes that occur in a human during one cardiac cycle.



When does systole start and end in the right atrium and in the right ventricle?

	in right atrium		in right ventricle	
	systole starts	systole ends	systole starts	systole ends
A	T	V	V	X
B	T	U	V	Y
C	U	V	W	X
D	V	W	W	Y

10. Nov/2023/Paper_ 9700/13/No.28

Which process can be carried out by a mature red blood cell?

A active transport

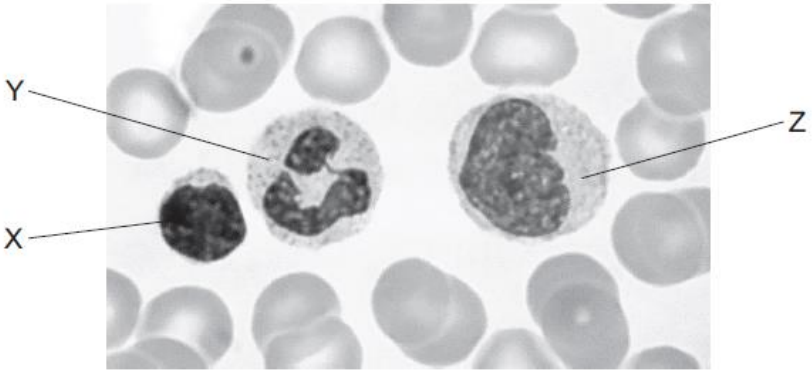
B cell division

C phagocytosis

D protein synthesis

11. Nov/2023/Paper_ 9700/13/No.29

The photomicrograph shows three white blood cells labelled X, Y and Z.



Which row correctly identifies these cells?

	cell X	cell Y	cell Z
A	lymphocyte	monocyte	neutrophil
B	lymphocyte	neutrophil	monocyte
C	monocyte	neutrophil	lymphocyte
D	neutrophil	monocyte	lymphocyte

12. Nov/2023/Paper_ 9700/13/No.30

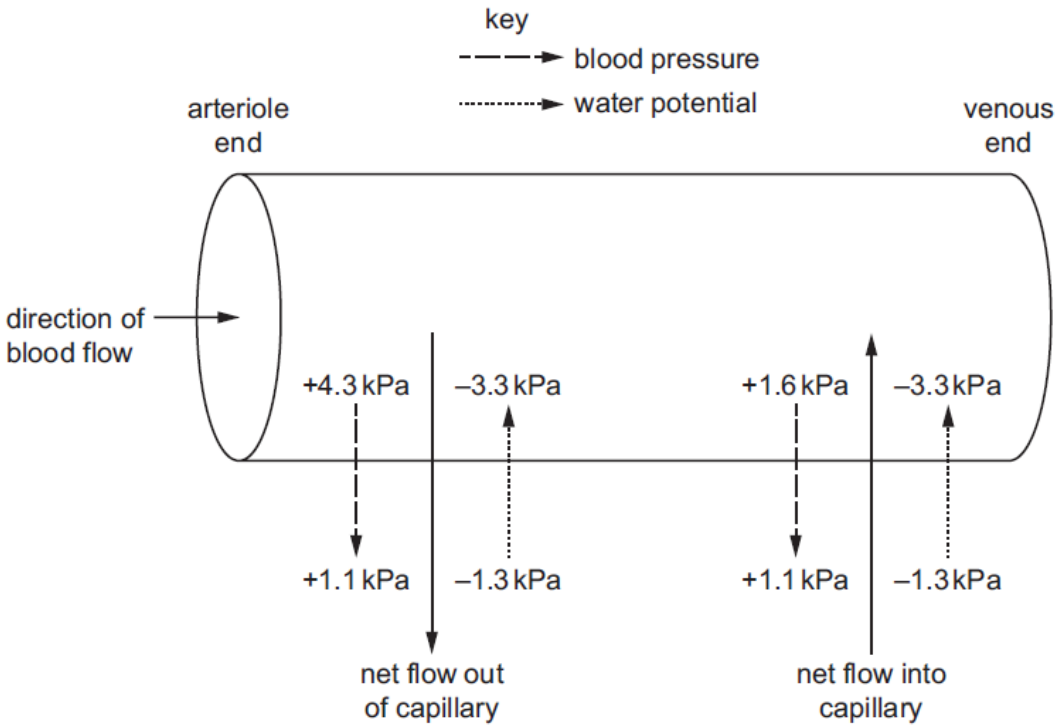
When a blood vessel is damaged it can result in problems in the body.

Which row identifies the blood vessel that could have been damaged?

	blood is not being distributed to organs that need it	blood is not returning to the heart and is collecting in organs	blood surges mean that relatively constant blood pressure is not maintained
A	elastic artery	muscular artery	vein
B	elastic artery	vein	muscular artery
C	muscular artery	elastic artery	vein
D	muscular artery	vein	elastic artery

13. Nov/2023/Paper_ 9700/13/No.31

The diagram shows how tissue fluid is formed.



What is the net pressure causing tissue fluid to flow out of the capillary at the arteriole end?

A –1.5 kPa **B** +1.0 kPa **C** +1.2 kPa **D** +5.2 kPa

14. Nov/2023/Paper_ 9700/13/No.32

What happens when carbon dioxide is transported by red blood cells?

A Hydrogencarbonate ions move into the red blood cells and chloride ions move into the plasma.

B Chloride ions move into the red blood cells and bind to haemoglobin.

C Chloride ions move into the red blood cells and hydrogen ions move into the plasma.

D Chloride ions move into the red blood cells and hydrogencarbonate ions move into the plasma.

15. Nov/2023/Paper_ 9700/22/No.3

The liver receives blood from the hepatic artery and from the hepatic portal vein. The hepatic portal vein transports blood from the digestive system.

Hepatocytes are the main cell type of the liver. They have a wide range of functions, including:

- the synthesis of triglycerides and plasma proteins
- detoxifying waste
- energy storage.

(a) The hepatic artery branches from the main artery that transports blood from the heart.

Name the main artery that transports blood to the hepatic artery.

..... [1]

(b) Blood arriving at the liver enters specialised blood vessels known as sinusoids.

Fig. 3.1 is a diagram of part of a sinusoid and surrounding hepatocytes. A second type of cell found in the liver, a Kupffer cell, is also shown.

Kupffer cells are phagocytic cells of the immune system.

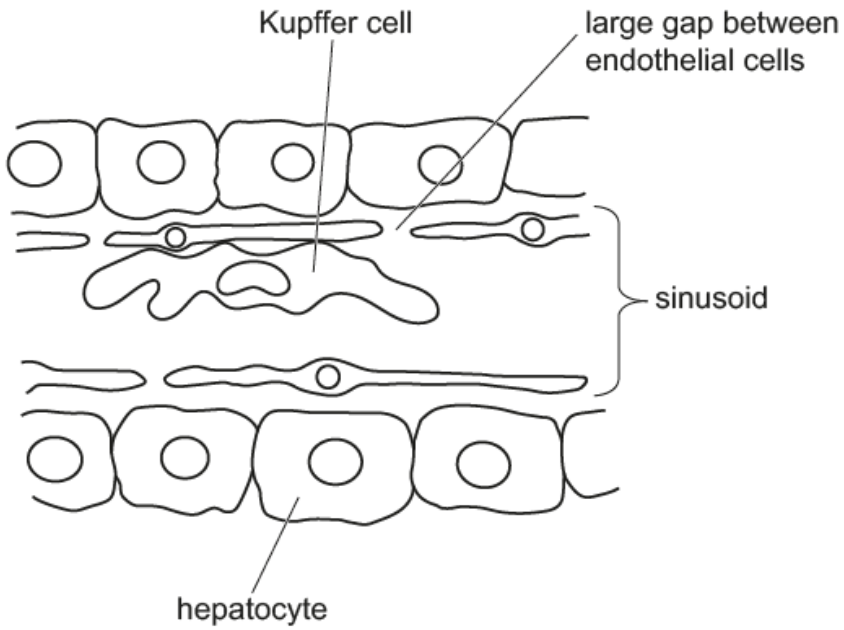


Fig. 3.1

(i) Suggest **one** advantage of having large gaps between the endothelial cells of the sinusoid, as shown in Fig. 3.1.

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..... [1]

(ii) In addition to removing bacteria present in the blood inside the sinusoid, Kupffer cells are also able to remove old or damaged red blood cells.

Describe the mode of action of a Kupffer cell in removing and breaking down a damaged red blood cell.

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..... [4]

(c) Fig. 3.2 is a transmission electron micrograph of part of a hepatocyte showing some cell structures.

The peroxisome shown in Fig. 3.2 is a spherical organelle bound by a single membrane. It carries out a variety of enzyme-catalysed metabolic reactions, including detoxification. Some of these reactions require oxygen.

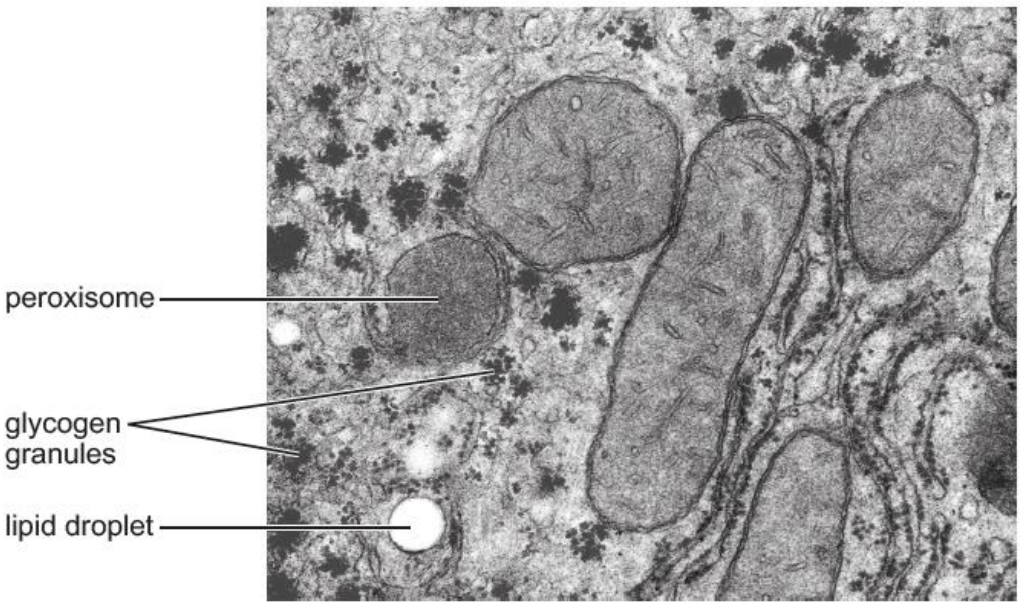


Fig. 3.2

(i) Describe the evidence **visible** in Fig. 3.2, apart from the presence of a peroxisome, that indicates some of the functions of a hepatocyte.

Add labels to Fig. 3.2 to identify the location of any cell structures, if not already labelled, that are part of your evidence.

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..... [3]

(ii) The mitochondria in Fig. 3.2 are larger than the peroxisome.

State **one other** difference, **visible** in Fig. 3.2, between a peroxisome and a mitochondrion.

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..... [1]

(iii) Some of the enzymes used within mitochondria can be synthesised by the organelle. Peroxisomes cannot synthesise any of the enzymes that they contain.

Suggest why a mitochondrion can synthesise enzymes, but a peroxisome cannot synthesise enzymes.

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..... [2]

(iv) One of the enzymes present in peroxisomes is catalase. This enzyme catalyses the breakdown of hydrogen peroxide to harmless products.

Suggest why it is useful to the cell for this reaction to take place within peroxisomes.

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..... [2]

[Total: 14]

16. Nov/2023/Paper_ 9700/23/No.6

The mammalian heart consists of four chambers.

(a) Explain why the ventricles have thicker walls than the atria.

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..... [2]

(b) Complete Table 6.1 to identify the structure in the heart responsible for each function listed.

Table 6.1

function	structure in the heart
initiates the heartbeat	
delays the wave of depolarisation between the atria and the ventricles	
transmits the wave of depolarisation through muscles of the ventricle	
closes when the left ventricle contracts	

[4]

[Total: 6]